

In-Space CanSat Competition

Context

The problem statement was to build a fully autonomous CanSat under 1Kg which will be launched to an altitude of 1Km and the CanSat must be equipped with two descent control mechanisms that have to be deployed automatically based on the altitude, while its descent the CanSat must relay live telemetry data to the ground station. The primary descent mechanism must be a parachute, the secondary can be any innovative mechanism. Our Team consisted of 6 members.

One of the main objectives of the competition was to have an “innovative” secondary descent control mechanism. Almost all teams went with the easy route - use a second parachute. But we wanted to push our boundaries and tried to incorporate a quad-copter system with deployable rotors, live telemetry, battery pack capable of providing peak current of 40 A , can last upto 2hrs on stand by mode and all of this inside a structure that was built for withstanding rocket launch and impact loads and still weighed less than 1Kg.

Scope and Contribution

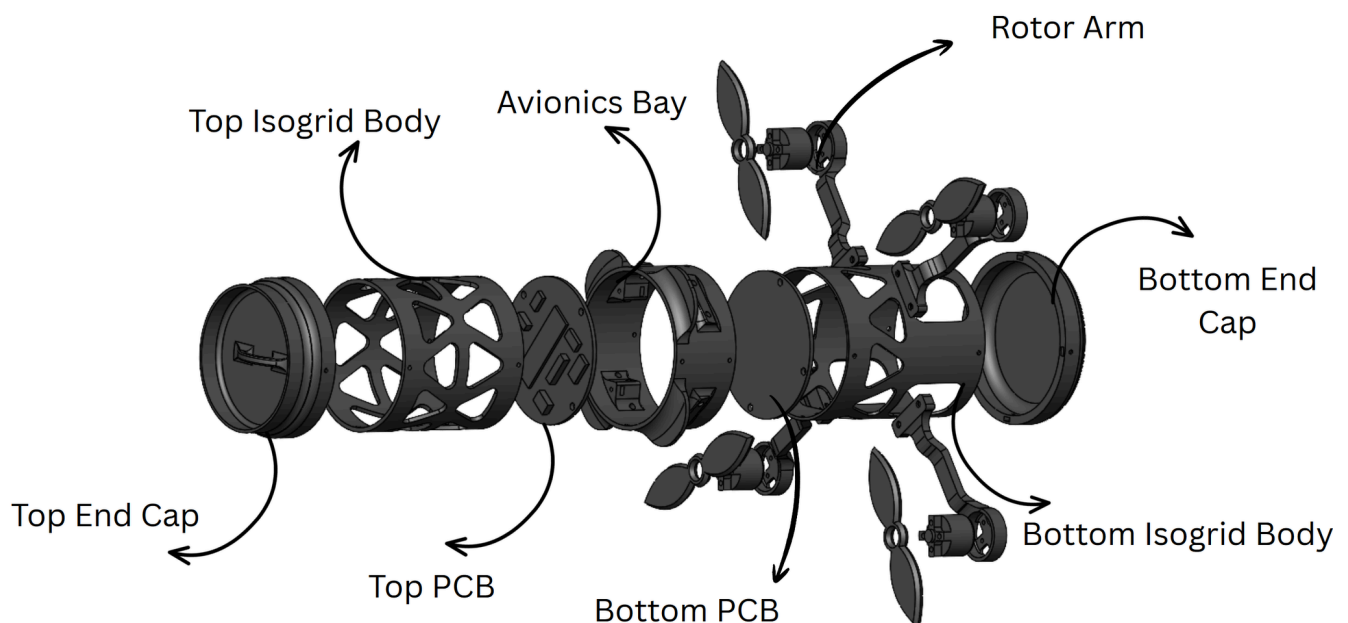
Being the only one in the Team to have a mechanical background, It was my job to lead and work on the entire structural design of the CanSat. Apart from that, I also contributed in building the control system for the secondary descent mechanism.

Technical Approach

Structural Design

The entire CanSat was designed in SOLIDWORKS and 3d printed in-house using ABS Resin.

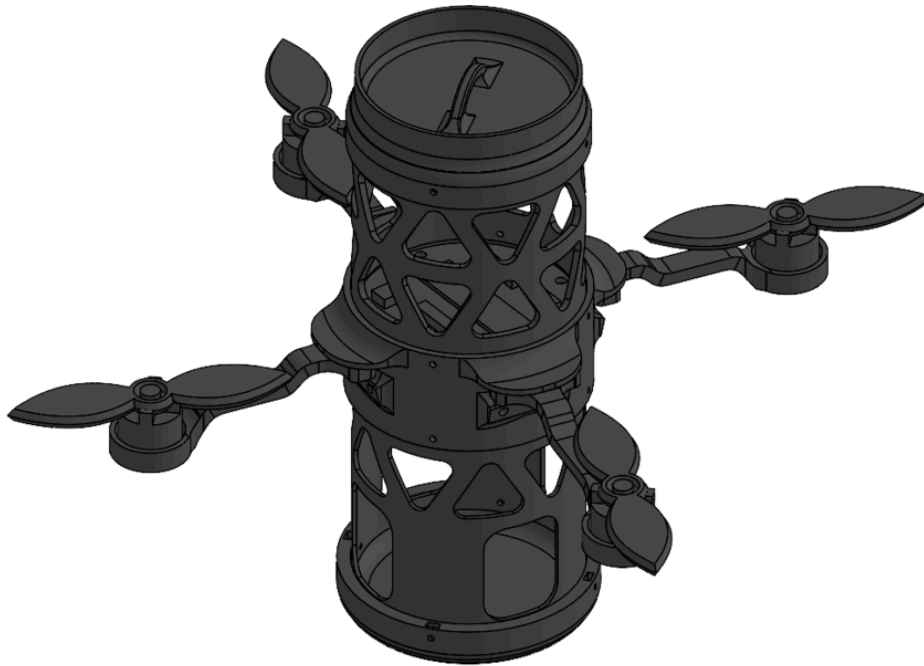
I chose a 5 piece modular design as opposed to a monolithic structure because even though the monolithic structure is easier to manufacture and would still weigh less, it had a huge disadvantage for internal access. This was very important for the team as we went through multiple changes, iterations and hours of debugging and required immediate access of the avionics and battery systems.



To reduce the overall weight of the structure I went with an isogrid main body which provided a good strength to weight ratio compared to a fully solid cylinder. The avionics bay was kept solid to house the deployment mechanism and the PCBs.

The entire structural design was done not only with Design for manufacturing principles but also keeping in mind the assembling of the structure. M3

fasteners were used throughout and the End caps contained holes for embedded nuts.



To achieve reliable deployment of the rotor, I chose a burn wire mechanism. The rotor arms were connected to the avionics bay using spring hinges with normal state as open. The arms were folded (appropriate space was given in the bottom body to accommodate the motors when the rotor was folded) and a Nylon wire was tied to keep them in closed position. A nichrome wire triggered by a relay was used to burn the Nylon wire and thus deploying the rotors.

The whole structure was simulated in ANSYS mechanical to ensure that it can withstand the launch and impact loads. A physical drop test was also conducted which further validated our design.

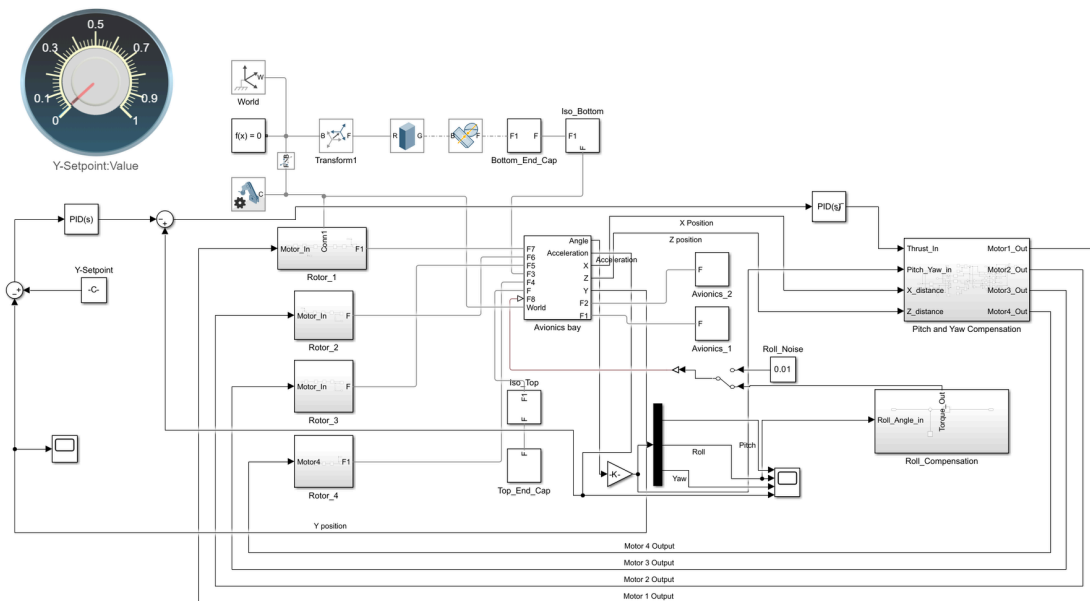
Analysis	Condition	Result
Shock, axial	295 m/s ² , 1ms impulse	0.01 MPa max
Shock, off-axis	axial+lateral	3.8MPa max (ABS UTS:36.5MPa)
Drop Test, explicit dynamics	3 m/s impact velocity	23MPa max, FoS=1.63
Modal	1st Mode	205 Hz
Rotor arm, static	240 Pa Dynamic Pressure +5 N motor Load	FoS=2.08
CFD, steady-state	20 m/s flow	Cd = 0.55 (bare structure)

Structural Simulation in ANSYS results

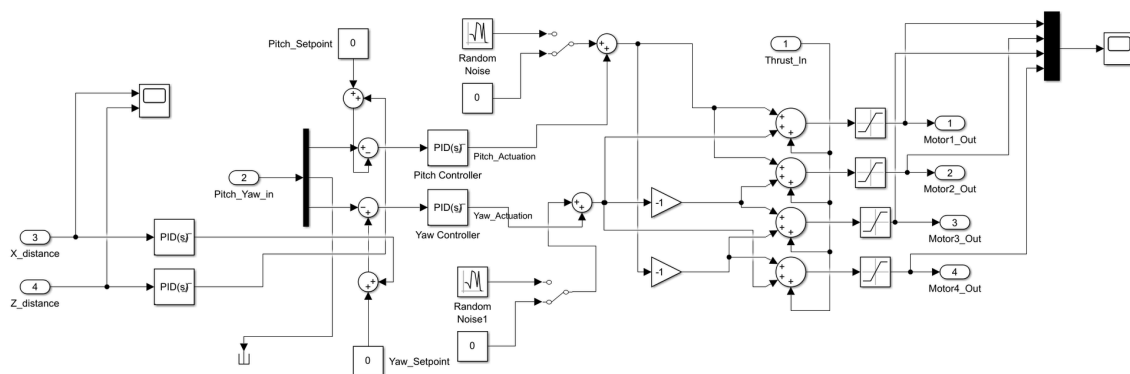
Control Systems

The main role of control systems in this build was the quad-copter control. Instead of using a readily available flight controller + ESC which would have made the job much easier, I chose to build the whole controller with Teensy from ground up. Before touching hardware, I built a simulink simulation where the CAD model was imported from SOLIDWORKS to simulink.

To incorporate the motors in the simulink model, I approximated them as an external Force+torque to the main body. I Started with building a single axis control and then extended it to Yaw,Pitch and Roll with motor mixing algorithm.



Apart from just maintaining attitude and decelerate during the descent, I also wanted to incorporate a precision landing feature so that the CanSat can land near the launchpad itself or land at any desired location. To implement this in simulink, I used a cascaded PID loop architecture to control all 6 DoFs of the CanSat.



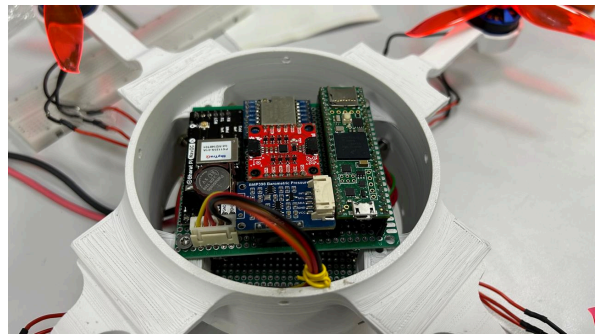
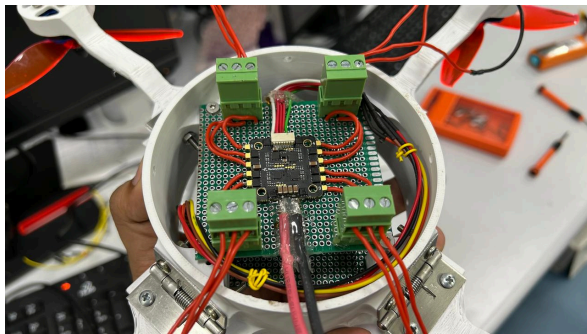
Later once I had the control algorithm working in simulink, I implemented the same in hardware.

Overview of Avionics

The Flight Computer chosen was Teensy 4.1. The avionics stack consisted of IMU (MPU6050), Pressure (BMP280) and temperature sensor, GPS, XBEE and 4-in-1 ESC for BLDC motor control.

The CanSat was powered by a custom Battery pack made with 6 18650 cells in 3S2P configuration. It was made sure that the cells have enough C-Rating to handle the current required by the motors.

Li-PO batteries were strictly prohibited in the competition which led to the use of Li-Ion cells. Charging was handled by a On-board BMS. ESC (Electronic Speed controller) had bulk electrolytic capacitors for handling current transients.

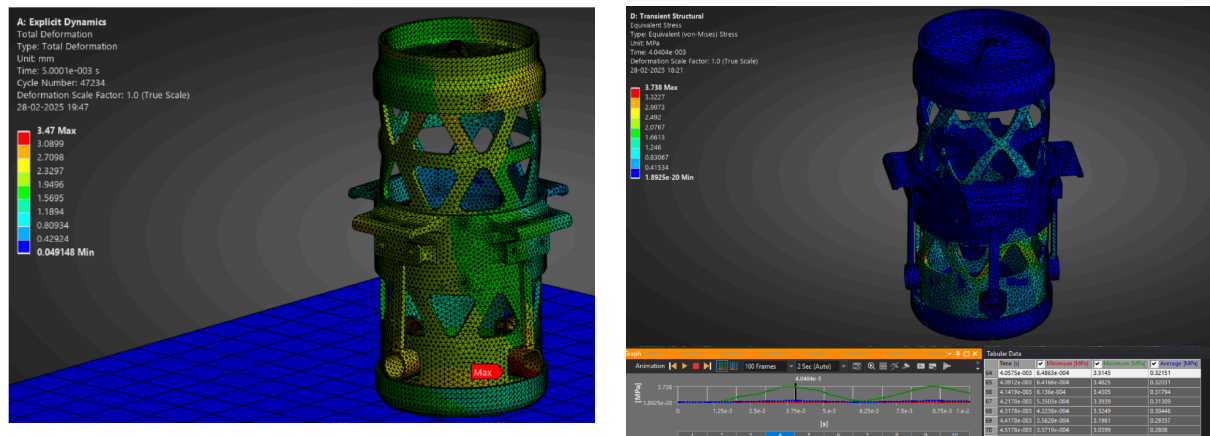


Testing, Iterations and Failures

Structures

- Started with multiple materials and design options, in the end chose ABS manufactured by SLA
- Tested two different manufacturing methods FDM vs SLA and chose SLA for good strength and reliability
- Simulated the structure under various launch and impact loads
- Since Spring hinges had a complex design and no CAD model of hinge was available, had to go through multiple prototypes to ensure proper mounting of spring hinge.
- Redesigned the rotor arm from a straight profile to a new design that matches the profile of the main body to keep the rotors as close to the

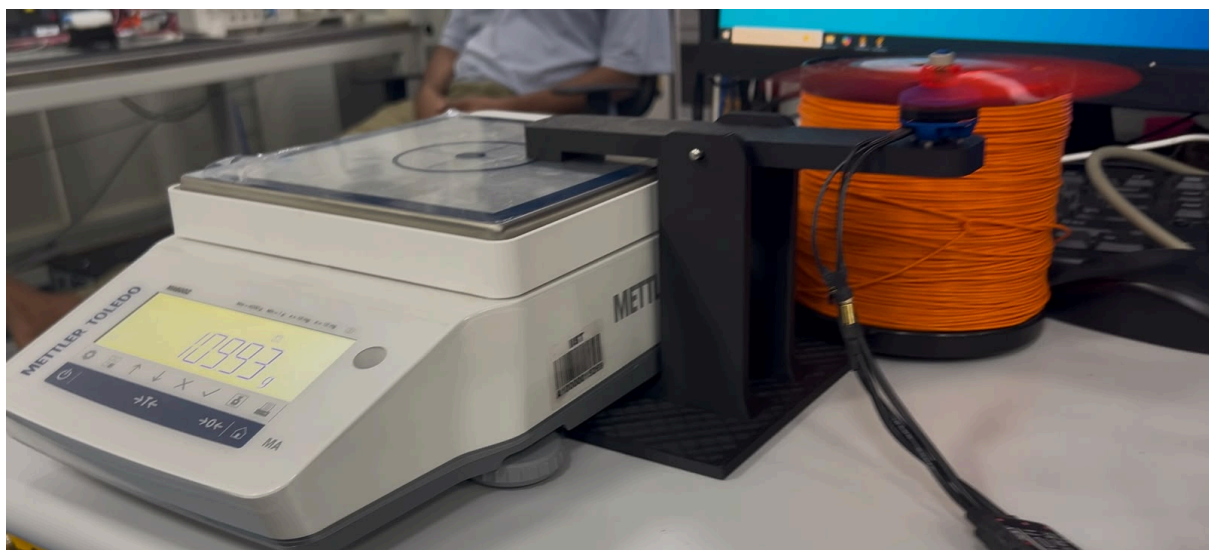
body as possible.



Impact analysis and Shock analysis

Control Systems

- Designed and 3d printed a test rig to measure the thrust characteristics of the motor



- Used the test rig to then implement a single axis control in hardware before moving to the CanSat
- Suspended the CanSat to test out the control algorithm, realized that the strings used for suspending provided some passive stability to the CanSat, but due to lack of time we could not build a better test rig.



Limitations and Remarks

- Although the 5 piece modular design was much better for debugging and provided easier access to internal components, it weighed heavier than a monolithic structure and also required additional fasteners which further increased the weight
- The test rig built for testing the control algorithm of the CanSat provided passive stability to the CanSat, although we tried to make its effect small by making sure the strings were not taught during the test, a better test rig is needed to properly validate the algorithm.
- Only a cascaded PID control was tested and implemented, future work could include looking into LQR as an alternative

Launch Outcome

The CanSat was launched successfully after clearing the jury and launch team review. The rocket was visible during the ascent, but due to bad weather and the CanSat being very small in size, it could not be visually tracked after separation from the rocket. Telemetry was also lost during that time. After clearance a 2hr search across the recovery fields also turned out unsuccessful as the CanSat could not be located. The rocket was found and CanSat separation was confirmed.

Onboard SD card logging was included by design specifically as a backup for scenarios where the live telemetry failed, and this redundancy worked as intended. However since the CanSat itself was never physically recovered, this backup provided no practical benefit, the logged data remains inaccessible without the hardware. The actual single point of failure was physical location and recovery of the unit. Future work should prioritize recovery aids like independent RF beacon, audible buzzer or standalone GPS tracker.